

GAINES COUNTY AP, TX, USA

WMO: 722107

Lat: 32.675N

Lon: 102.654W

Elev: 3315

StdP: 13.02

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 03056

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.7	22.8	2.2	6.9	36.0	5.8	8.4	41.5	29.1	54.3	26.1	56.2	6.3	40	0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.1	100.7	64.8	98.7	65.2	96.4	65.2	71.3	85.1	70.4	84.2	69.4	83.4	10.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.4	118.0	74.0	67.4	114.0	73.5	66.0	108.6	73.1	37.3	85.2	36.4	83.7	35.7	83.1	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.8	20.7	18.3	DB	12.5	105.9	4.6	2.7	9.2	107.8	6.5	109.3	4.0	110.8	0.6	112.7	(4)
(5)			WB	11.0	72.9	4.3	1.7	7.8	74.1	5.3	75.0	2.9	76.0	-0.3	77.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.7	43.1	47.2	55.4	62.7	70.7	80.0	80.6	80.0	73.0	62.9	51.9	43.7		
	DBStd	15.60	8.67	9.72	8.84	7.86	7.90	4.85	4.13	4.44	5.64	7.89	8.44	8.75		
	HDD50	769	244	149	49	9	1	0	0	0	0	10	77	230		
	HDD65	2860	679	499	312	131	38	0	1	0	9	132	398	661		
	CDD50	5393	30	71	215	389	644	901	949	929	689	409	133	34		
	CDD65	2007	0	1	13	61	216	451	484	464	247	67	3	0		
	CDH74	25203	11	86	448	1254	2966	5687	5799	5373	2515	901	148	15		
	CDH80	13099	0	13	113	479	1523	3301	3243	2957	1145	300	25	0		
(14) Wind	WSAvg	8.3	8.5	9.1	9.4	10.3	9.5	8.9	7.3	6.8	6.6	7.4	7.7	8.3		
(15) Precipitation	PrecAvg	17.8	0.6	0.7	0.9	0.9	2.1	2.3	2.4	2.2	2.5	1.4	0.9	0.6		
	PrecMax	33.2	2.3	2.4	4.5	3.6	7.6	10.2	9.8	7.5	9.2	5.6	6.2	3.6		
	PrecMin	7.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	5.5	0.6	0.7	1.0	1.1	1.8	1.9	2.2	1.9	2.4	1.6	1.2	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.3	82.0	87.4	92.4	100.4	105.4	102.0	102.2	97.2	90.4	83.6	76.6		
		MCWB	50.3	53.4	55.0	57.2	61.0	63.3	66.0	66.1	65.8	61.8	55.5	50.8		
	2%	DB	71.3	76.8	83.4	89.4	96.9	101.8	99.3	99.2	93.5	87.1	78.6	71.5		
		MCWB	48.7	51.3	54.5	56.1	60.1	63.6	66.5	66.9	64.8	61.5	54.5	50.0		
	5%	DB	65.7	72.2	79.4	86.0	93.2	99.0	97.0	96.7	90.6	83.4	73.6	65.9		
		MCWB	46.5	49.6	52.9	55.4	59.4	64.3	66.6	66.7	64.6	60.2	52.9	47.7		
	10%	DB	60.8	66.8	74.8	81.9	89.2	95.7	94.6	94.0	87.2	79.4	69.4	60.4		
		MCWB	44.5	47.3	51.6	55.1	59.4	64.8	66.7	67.0	64.6	59.5	51.4	45.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.5	56.3	62.1	64.7	69.2	71.4	73.2	72.9	71.5	69.4	60.7	55.7		
		MCDB	62.5	71.4	71.5	76.0	81.1	86.9	83.9	84.9	81.5	80.2	68.4	62.8		
	2%	WB	51.0	53.4	59.5	62.4	67.5	70.1	71.2	71.3	69.9	67.5	58.1	51.7		
		MCDB	65.2	69.7	69.2	73.0	78.1	85.7	84.4	86.2	80.3	76.9	68.2	65.9		
	5%	WB	48.1	51.0	57.0	60.2	65.9	69.2	70.0	70.3	68.8	65.3	55.8	49.5		
		MCDB	64.2	68.8	69.2	72.1	77.4	84.7	84.2	84.8	79.4	75.0	68.1	64.0		
	10%	WB	45.4	48.5	54.4	58.3	64.2	68.3	69.1	69.3	67.8	63.2	53.6	46.5		
		MCDB	60.1	65.4	70.0	71.7	76.6	83.8	84.0	83.5	78.6	72.1	66.5	59.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	27.0	28.2	29.8	30.6	28.0	27.9	25.1	25.3	23.9	26.7	27.9	25.9		
		MCDBR	36.0	37.1	37.0	36.3	34.5	32.6	29.0	29.0	30.0	31.9	34.5	35.5		
	5% WB	MCWBR	18.3	17.6	15.6	14.4	11.8	8.6	7.4	7.1	8.6	11.8	15.5	18.0		
		MCWBR	32.0	34.0	31.1	29.1	26.7	27.3	24.5	24.5	22.4	24.3	28.7	31.4		
(40) Clear-Sky Solar Irradiance	taub	taub	0.261	0.272	0.292	0.323	0.337	0.374	0.391	0.370	0.349	0.315	0.284	0.271		
		taud	2.571	2.543	2.499	2.404	2.429	2.377	2.372	2.402	2.434	2.506	2.527	2.538		
	Ebn at Noon	Ebn at Noon	310	316	314	306	300	288	282	288	291	295	298	299		
		Edh at Noon	25	29	32	37	37	39	39	37	34	30	26	25		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1106	1371	1775	2188	2343	2455	2317	2143	1785	1456	1154	974		
	RadStd	RadStd	105	125	131	133	160	116	120	90	128	148	117	80		

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page